

Birla Vishvakarma Mahavidyalaya
Engineering College
[An Autonomous Institution]

A
Project Report
On

**NotesHub – A Version-controlled Knowledge management
system**

Under the course of
DESIGN ENGINEERING -3CP08
B. E. Semester – VI
(Computer Engineering)

Submitted by:

Sr.	Name of student	ID No.
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Academic year
(2025-2026)

CERTIFICATE

This is to certify that the students namely,
Het Soni (23CP050) of ***B. E. (Computer Engineering)***
Semester VI have successfully completed the course work and related
tasks for the course of **Design Engineering 3CP08** during the academic
term ending in the month of April 2025.

Date: 10/04/26

Faculty Signature

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Date: 10/04/26

Faculty Signature

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ABSTRACT

In today's academic world, students organize their study resources through various platforms like messaging apps, PDF documents, online resources, and even physical notepads. This causes fragmentation, problems with revising, and discontinuity of knowledge.

NotesHub is an academic notes management system that utilizes Git's principles for version control. NotesHub uses snapshotting, restore feature, and note linkage to form a systematic knowledge base. The application guarantees security during the editing process and keeps track of all changes made to maintain the development journey of the user's learning process.

Spring Boot and PostgreSQL are used for implementing the backend of the NotesHub. The project emphasizes the implementation of design engineering in practical case

CHAPTER 1

INTRODUCTION

Problem Statement

The students have problems with organizing the information in the following ways:

- Separation of notes in many places
- Absence of version control
- Overwriting of crucial information
- Revisions are hard to do
- Not having links between relevant information

It leads to ineffective learning process.

Project Objectives

The goals of NotesHub include:

- The design of an integrated notetaking system
- Version control to manage educational notations
- The provision of restore feature for secure editing
- Conceptual linkage capability
- Scalability and modularity of the backend
- Revision enhancement and knowledge management

CHAPTER 2

LITERATURE REVIEW AND PROJECT METHODOLOGY

Literature Review

There already exist tools like Notion, OneNote, and Obsidian that offer note-taking functionality but fail to offer:

- Version History Structure
- Restore Functionality
- Academic-Focused Features
- Linking of Concepts

Git, which is a version control system, uses snapshots to maintain history. The current project adopts a similar approach for academic purposes.

Methodology

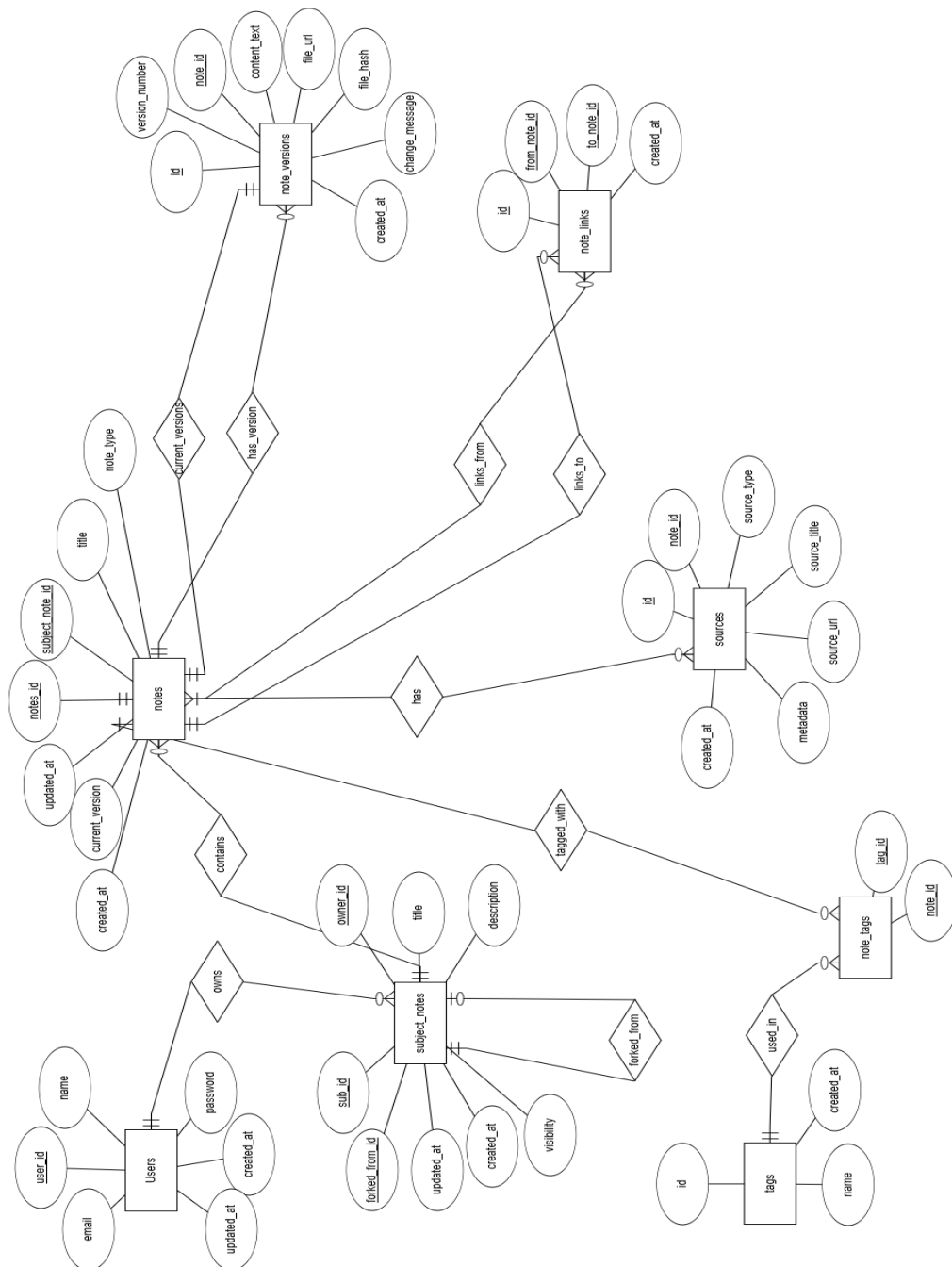
The design of this system is done based on Design Engineering methodology, which includes:

- Problem Identification
- User Analysis
- Ideation
- System Design
- Implementation
- Testing & Evaluation

CHAPTER 3

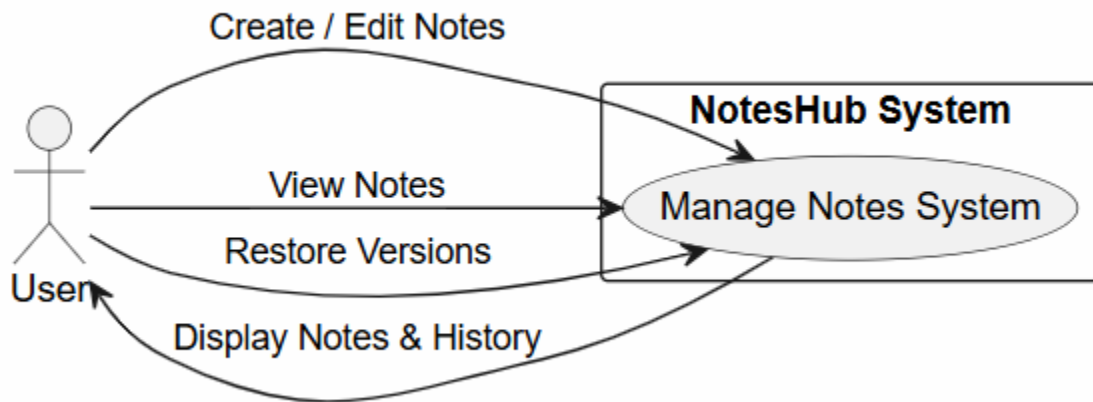
Design Considerations For Detail Design

NotesHub ER Diagram

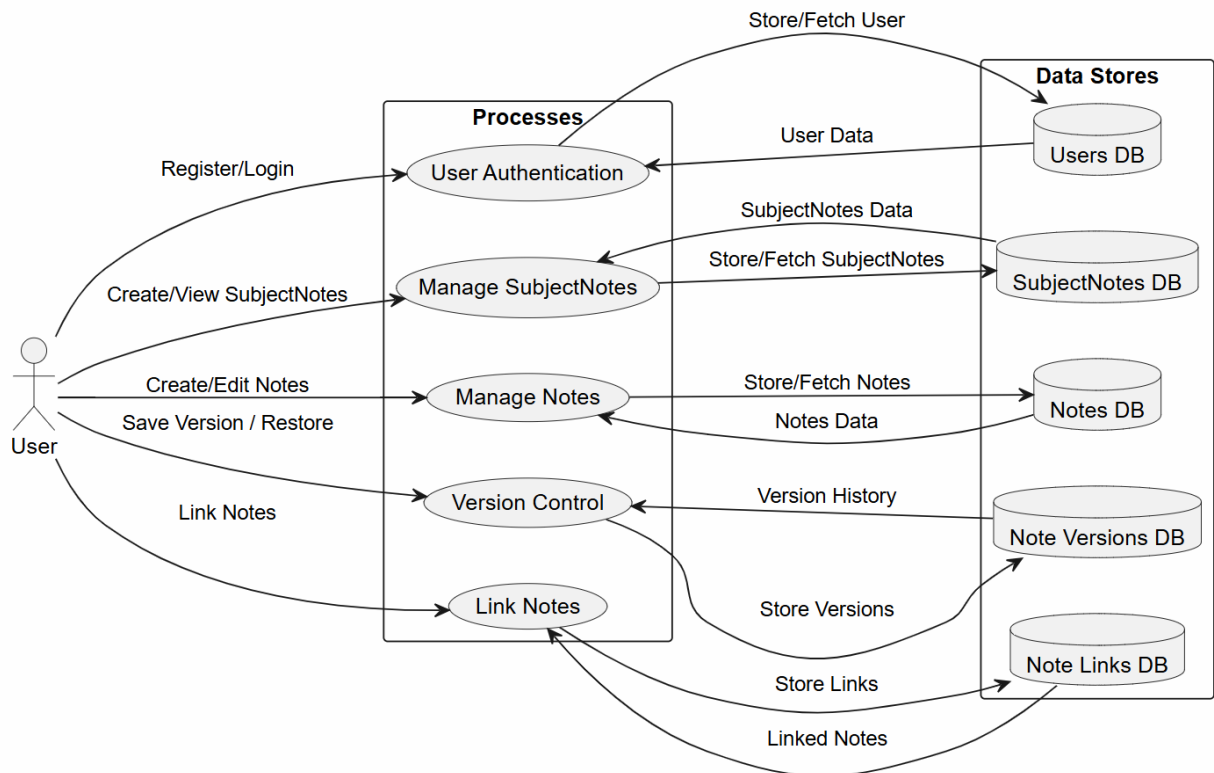


DFD diagram

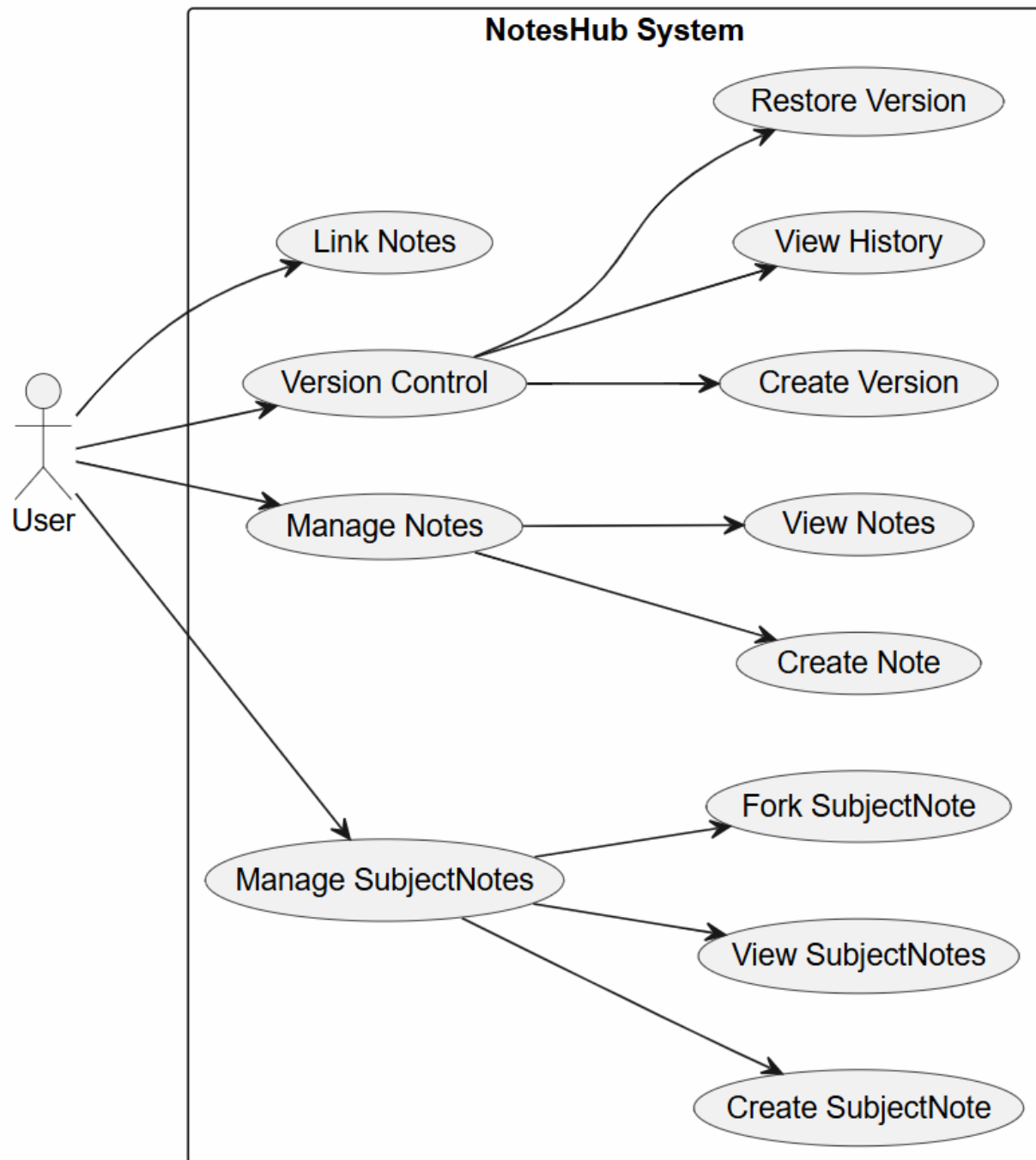
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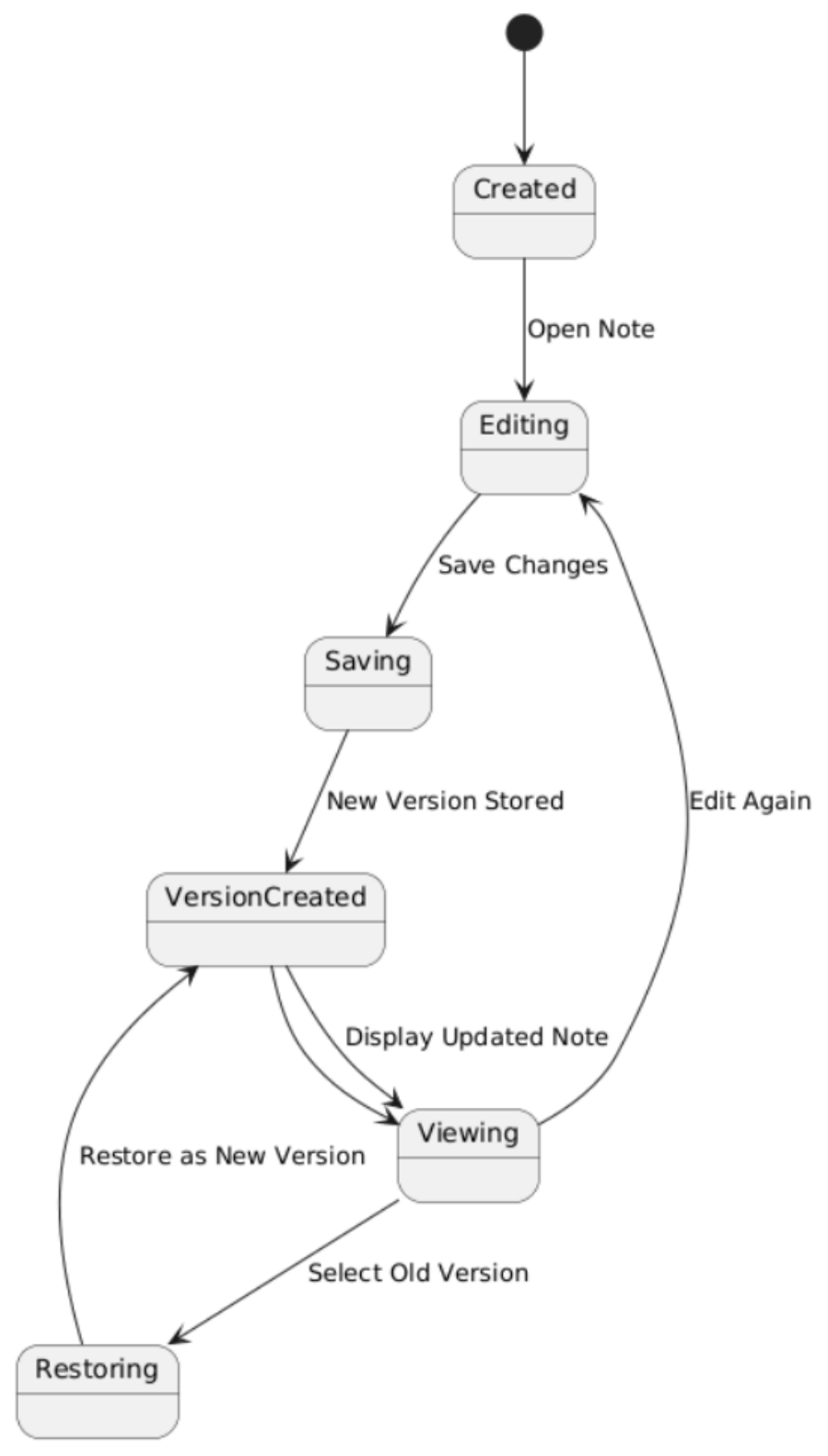
Level 1



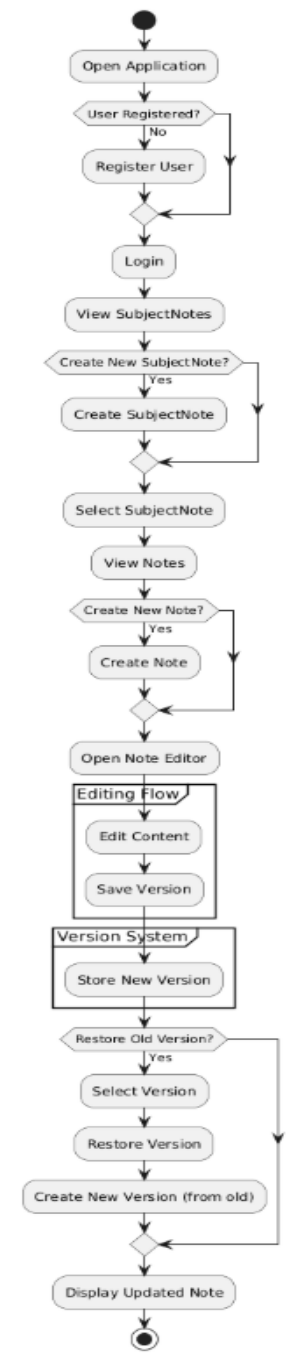
Use Case diagram



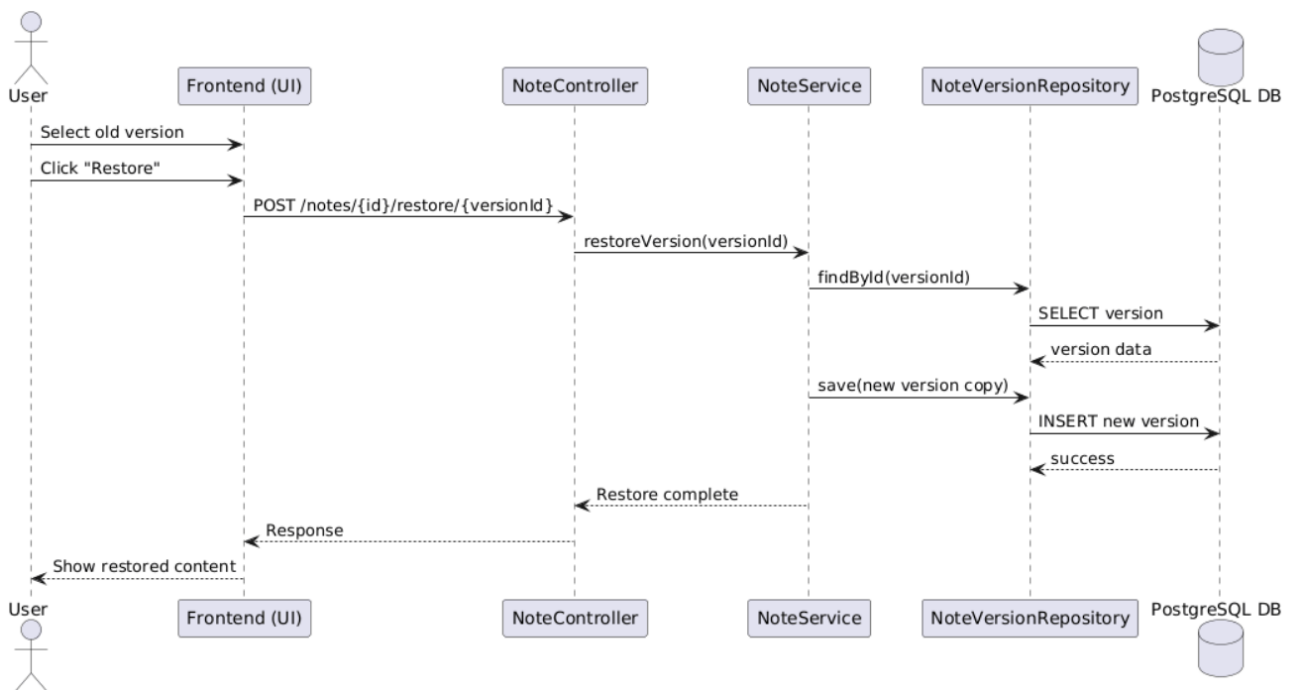
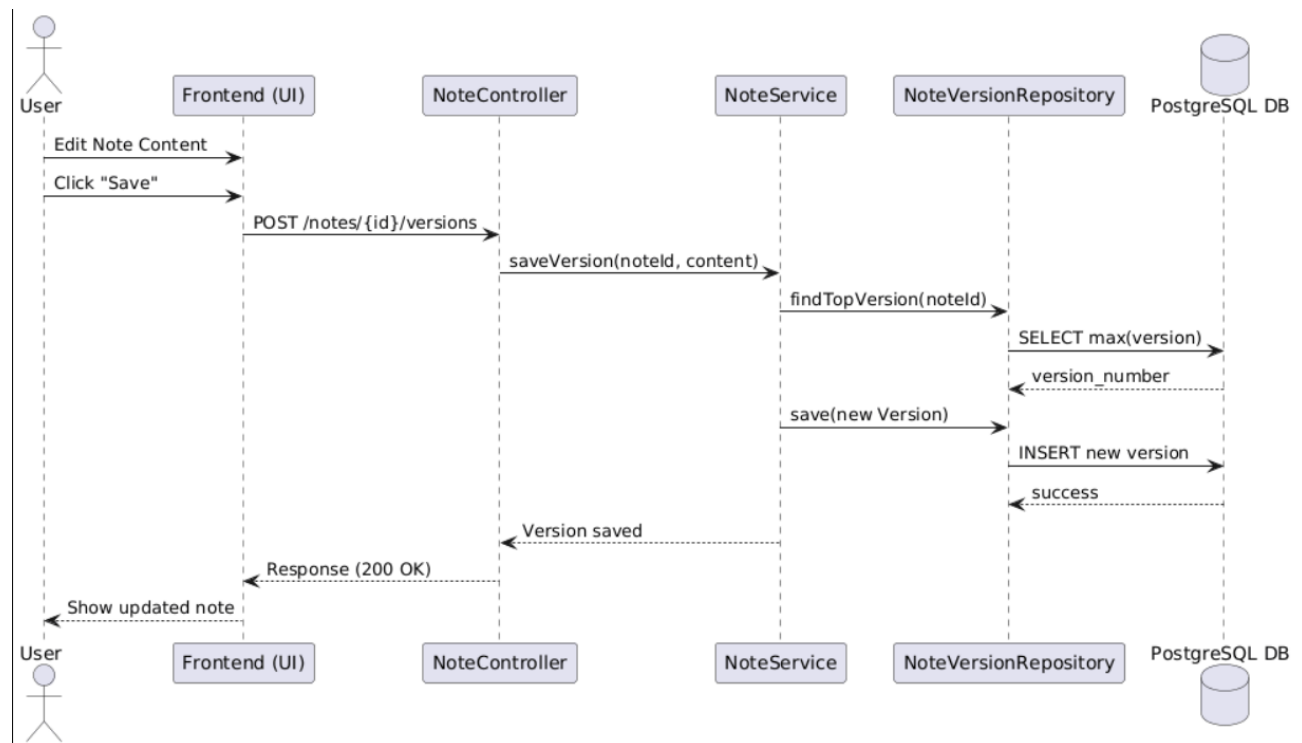
State Diagram



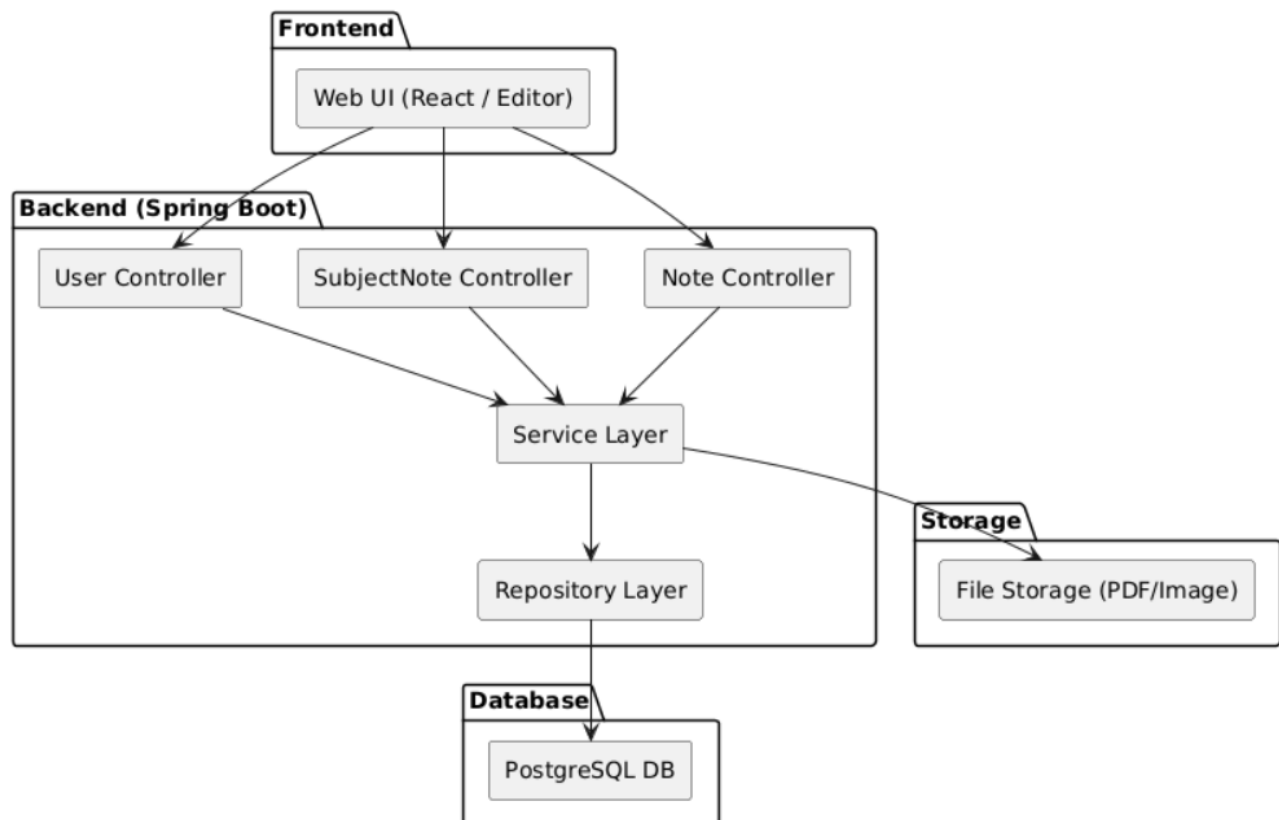
Activity Diagram



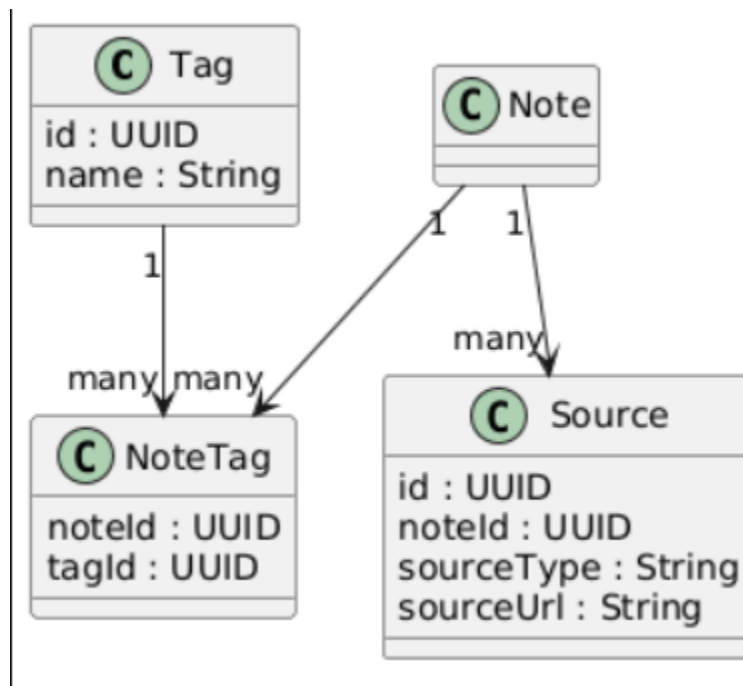
Sequence Diagram



Component Diagram



Class Diagram



CHAPTER 4

CANVASES

AEIOU :

AEIOU Summary		Group ID:	Date: 10-04-2026	Version: 1.0
Domain Name:				
Environment :	Interaction :	Objects :		
• Academic Environment • Digital learning platforms	• User interacting with system • Backend storing versions	• Notes • PDFs • Devices		
Activities :		Users :		
• Creating notes • Editing notes • Revising content • Linking concepts		• Students • Faculty		

EMPATHY :

Design for : NotesHub		Design By : Meet and Het	
Date: 10-04-2026		Version : 1.0	
USER • Students • Faculty		STACKHOLDER • Educational Institutions	
ACTIVITIES • Studying • Revising • Organizing Notes			
STORY BOARDING HAPPY: Easy Note Creation A student logs into NotesHub and creates a new SubjectNote for "Operating Systems". They quickly add notes from their lecture and save them. The system automatically creates a version of the note so the student knows their work is safely stored.			
HAPPY : Restoring an older version While editing notes, the student accidentally removes an important explanation. Using the version history feature, they open the previous version and restore it. Outcome: The student easily recovers lost information and continues studying without hesitation.			
SAD: Disorganized notes without system Before using NotesHub, the student stores notes in multiple places such as Whatsapp chats, PDFs and notebooks. When preparing for exams, they struggle to find information and waste time searching across different platforms. The student feels frustrated due to lack of organization.			
SAD: Accidental Data Loss A student edits a document in a traditional note-taking app and mistakenly deletes important content. Since the app does not maintain version history, the student cannot recover the lost information. The student loses important notes and becomes stressed before exams.			

IDEATION CANVAS :

The ideanaut : <i>Ideation Canvas</i>		Project : NotesHub	Team :
 People Students Developers			
 Activity - Note Creation - Version Tracking - Revision	 Situation/Context/Location - Home - College		
 Props/Possible Solution - Develop NotesHub, a centralized note management system for students. - Organize notes using subject-based containers (SubjectNotes). - Implement version control so every edit creates a new version of the note. - Provide restore functionality to recover previous version.			

PRODUCT DEVELOPMENT :

Product Development Canvas		Team :	Date : 10-04-2026	Version : 1.0
 Purpose_Info • To provide a centralized platform for managing academic notes • To prevent loss of information through version control • To improve organization and revision efficiency for students • To enable structured knowledge management	 Product Experience • Simple and clean interface • Organized notes by subjects and topics • Easy access to version history and restore options • Smooth note editing & navigation	 Customer Revaldation • Students want organized notes in one place • Need ability to recover deleted content • Require simple and fast note editing • Prefer structured revision tools		
	 Product Function • Create and manage Subject Notes • Create and edit notes within subjects • Maintain version history of notes • Restore previous versions • Link related notes together			
	 Product Features • Version controlled notes • Restore old versions • Subject-based organization • Note linking for related concepts • Secure backend storage			
 People_Info • Primary users: Students • Secondary users: Teachers / Learners • Developers: Project Team • Stakeholders: Educational institutions	 Component_Info • Frontend: Web interface / note editor • Backend: Spring Boot REST API • Database: PostgreSQL • Storage: Note content and version history	 Reject, Redesign, Retain • Reject: Traditional note apps without version tracking • Redesign: Improve UI for easier note navigation • Retain: Version control system Subject-based note organization		

Learning need matrix : NotesHub

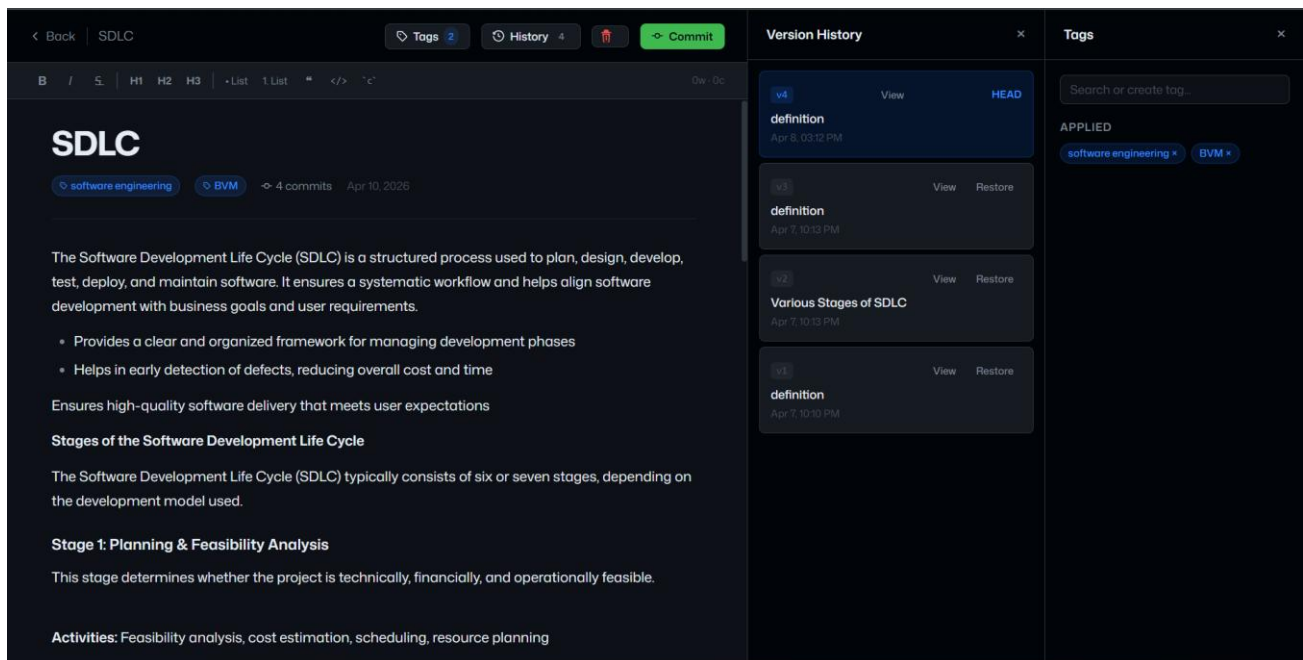
Theories/ Methods/ Application Process Involved/ Mathematical Requirement	During BE III / Stage III		Applicable standards and design specifications/ Principles & Experiments
• Authentication Concepts • UI/UX Principles	During BE III / Stage II		• Security Guidelines • Deployment Standards
	• Cloud Architecture • Access Control Logic	• Secure Data Handling • REST API Standards	
	During BE III / Stage II		
	• Auth Concepts • UI/UX Principles	• API Commn. Basics	
	Domain of Skill-sets to be learned Spring Boot Backend Development REST API Design Version Control Logic Database Integration		
• Spring Boot Setup • PostgreSQL Basics • Git Version Control	• Spring Boot Setup • PostgreSQL Basics	• API Testing • Database Testing	• Scale Eng Testing • Deployment Tools
	• Backend API Development • Database Integration • Version Management	• Integration Testing • System Testing	
• Spring Boot Setup • PostgreSQL Basics • Git Version Control		• System Performance Testing	• Cloud Deployment Tools • Performance Optimize
Software/ Tools/ Simulation Methods/ Skill			Component materials & strength criteria (exploration – varieties/testing requirems)

CHAPTER 5

IMPLEMENTATION

Entire implementation code for project is available in the following link :

<https://github.com/sonihet50/NotesHub>



Notes Editor

Dashboard

NotesHub

Dashboard

My Notebooks

Explore

Bookmarks

Welcome back, het

Your git-inspired note management dashboard

4Notebooks

3Public

2Forked

QUICK ACTIONS

View My Notebooks

RECENT NOTEBOOKS

Software EngineeringSDLCpublic

Machine LearningAbout machine learning algorithmprivate

Theory of Computation (fork)About Theory of Computationpublic

Explore

Discover public notebooks created by the community. Fork them to your account to read and modify.

Search public notebooks by title, description, or tags...

Software Engineering (fork) (fork)het

SDLC

Last updated 4/9/2026

Bookmark

Fork

Software Engineering (fork)meet

SDLC

Last updated 4/8/2026

Bookmark

Fork

Theory of Computationmeet

About Theory of Computation

Last updated 4/7/2026

Bookmark

Fork

Software Engineeringhet

SDLC

Last updated 4/7/2026

Bookmark

Fork

22

Notes List

Notebooks > Software Engineering

Software Engineering

+ New Note

1 note

TITLE	TYPE	VERSION	CREATED
☐ SDLC	TEXT	v4	Apr 7 

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Conclusion

NotesHub transforms traditional note-taking into an evolving knowledge system. By applying version control principles, it ensures safe editing, structured revision, and knowledge continuity. The system demonstrates how software engineering concepts can be applied to real-world academic problems.

Future Scope

- Tag-based organization
- AI-based summarization
- Collaborative editing
- Advanced search functionality
- Graph-based visualization

CHAPTER 7

REFERENCES

- Git Documentation – <https://git-scm.com>
- Spring Boot Documentation – <https://spring.io>
- PostgreSQL Documentation – <https://postgresql.org>
- React Documentation – <https://react.dev/learn>
- Vanilla CSS Documentation – <https://vanillaframework.io/>
- Design Engineering Concepts